

Half-Horizon Delayed Reset Suffixes

Finite-Prefix Invariance and a Sharp Retention–Conditioning Frontier

Prime Dynamics Theory Program

July 2026

Abstract

RH-153 showed that correlated reset Gram–tail pairs avoid quadratic inverse-overlap loss, but the full frozen atlas still contains one robust overlap as small as 8.9866×10^{-5} and a maximum log conditioning drawdown of 32.015. These extremes are concentrated in finite birth prefixes.

We prove that deleting finitely many reset transitions preserves an eventual positive support liminf and the existence of a lower-bounded support-cocycle drawdown. For a finite family of chains we then solve the exact optimization problem: among terminal suffixes retaining at least a fraction q of every chain, the largest common overlap floor is the minimum over the last $\lceil qn_j \rceil$ transitions of each chain. No search or greedy rule is needed, and the bound is sharp.

At $q = 1/2$, the frozen atlas retains 62 of 120 transitions. Its common robust overlap rises to 0.0666397, the inverse upper falls from 11127.6 to 15.0061, and the maximum log drawdown falls from 32.0150 to 8.9730. The minimum correlated transported base remains positive at 2.2304×10^{-7} . One may retain at least 58.8% of every chain with common overlap 0.05. Thus a delayed reset removes the finite conditioning spike without weakening the frozen base floor. It does not prove that the same fractional suffix is uniformly conditioned at untested levels.

1 Why a delayed reset is legitimate

The target statement in the present route is eventual positive directional support. If s_n denotes the support quantity, it asks for

$$\liminf_{n \rightarrow \infty} s_n > 0.$$

Likewise, the correlated support cocycle is controlled by lower bounds on partial sums of logarithmic multipliers. Neither property depends on a finite initial segment. This permits a reset after a finite birth layer, provided the new starting pair is itself certified. We use only elementary order and logarithmic inequalities in this reduction [Hardy et al., 1952, Horn and Johnson, 2013].

Theorem 1 (Finite-prefix invariance). *For every finite N ,*

$$\liminf_{n \rightarrow \infty} s_n = \liminf_{n \rightarrow \infty} s_{n+N}.$$

If ℓ_n is a real cocycle increment sequence, then the existence of N and $K < \infty$ such that

$$\sum_{j=N}^m \ell_j \geq -K \quad (m \geq N)$$

is unchanged by deleting or adjoining finitely many increments.

Proof. Deleting finitely many terms does not change the set of subsequential tail limits. For the cocycle, changing the starting index adds one fixed finite sum to every later partial sum; this only changes the lower-bound constant. $\square$

The theorem does not allow one to skip infinitely many bad transitions. A delayed route still requires an all-level law excluding recurrent birth walls.

2 Sharp retention–conditioning optimization

Consider positive overlap lowers $a_{j,1}, \dots, a_{j,n_j}$ on finitely many chains. A delayed reset keeps a terminal suffix because all later transitions must remain composable.

Theorem 2 (Optimal fixed-retention suffix). *Fix $q \in (0, 1]$ and put $L_j = \lceil qn_j \rceil$. Among all terminal suffixes of chain j containing at least L_j transitions, the largest possible floor is*

$$A_j(q) = \min_{n_j - L_j < t \leq n_j} a_{j,t}.$$

For the entire family, the sharp common floor is

$$A(q) = \min_j A_j(q).$$

The corresponding inverse upper is $A(q)^{-1}$, and the largest retained log drawdown is

$$\max_j \sum_{t=n_j-L_j+1}^{n_j} -\log a_{j,t}.$$

Proof. A terminal suffix with at least L_j entries contains the last L_j entries. Its minimum can therefore be no larger than $A_j(q)$. Keeping exactly those last L_j entries attains $A_j(q)$. Taking the minimum over chains gives the common optimum; the inverse and logarithmic formulas are identities. $\square$

This extremal statement is elementary but useful: it prevents selecting a visually favorable delayed start that quietly discards too much of one channel.

3 The terminal-half certificate

The ten RH-153 chains have lengths 4, 4, 6, 6, 11, 11, 17, 17, 22, 22. Keeping their terminal halves therefore retains

$$2 + 2 + 3 + 3 + 6 + 6 + 9 + 9 + 11 + 11 = 62$$

transitions. The sharp common overlap floor is

$$A(1/2) = 0.06663973169,$$

attained on the $\sigma = 0.02$ right channel. Thus every retained inverse overlap has norm at most 15.0061. The maximum chain drawdown is 8.9730, compared with 32.0150 on the full atlas.

The delayed suffix does not improve the minimum correlated base, because that minimum occurs near the terminal weak mode rather than in an overlap birth spike. Importantly, it does not worsen it either: all 62 retained base lowers are positive and their minimum remains 2.2304×10^{-7} . Hence the conditioning and weak-base mechanisms are separated. Prefix deletion repairs the former; the latter requires a native Gram–tail/support argument.

The full retention frontier supplies nearby alternatives. Requiring common overlap at least 0.05 permits suffixes retaining at least $10/17 = 58.8\%$ of every chain, 99 transitions in total. Requiring 0.1 is much more expensive because the $\sigma = 0.02$ right chain has a late overlap valley.

4 Composition consequence and boundary

Suppose a future source theorem certifies one reset Gram–tail pair at the beginning of a terminal suffix. RH-153 transports its relative tail without inverse loss; Theorem 2 supplies a finite uniform overlap floor and base factor on the frozen half-horizon. Theorem 1 then shows that omitted births do not alter an eventual conclusion.

What remains missing is the native reset tail pair, an all-level theorem that a fixed fractional suffix remains well conditioned, and a lower-bounded support cocycle across scales. We have not established Stage A, constructed a Hilbert–Polya operator, identified zeta zeros, or proved the Riemann Hypothesis.

References

- G. H. Hardy, J. E. Littlewood, and G. Polya. *Inequalities*. Cambridge University Press, second edition, 1952.
- Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. Cambridge University Press, second edition, 2013.

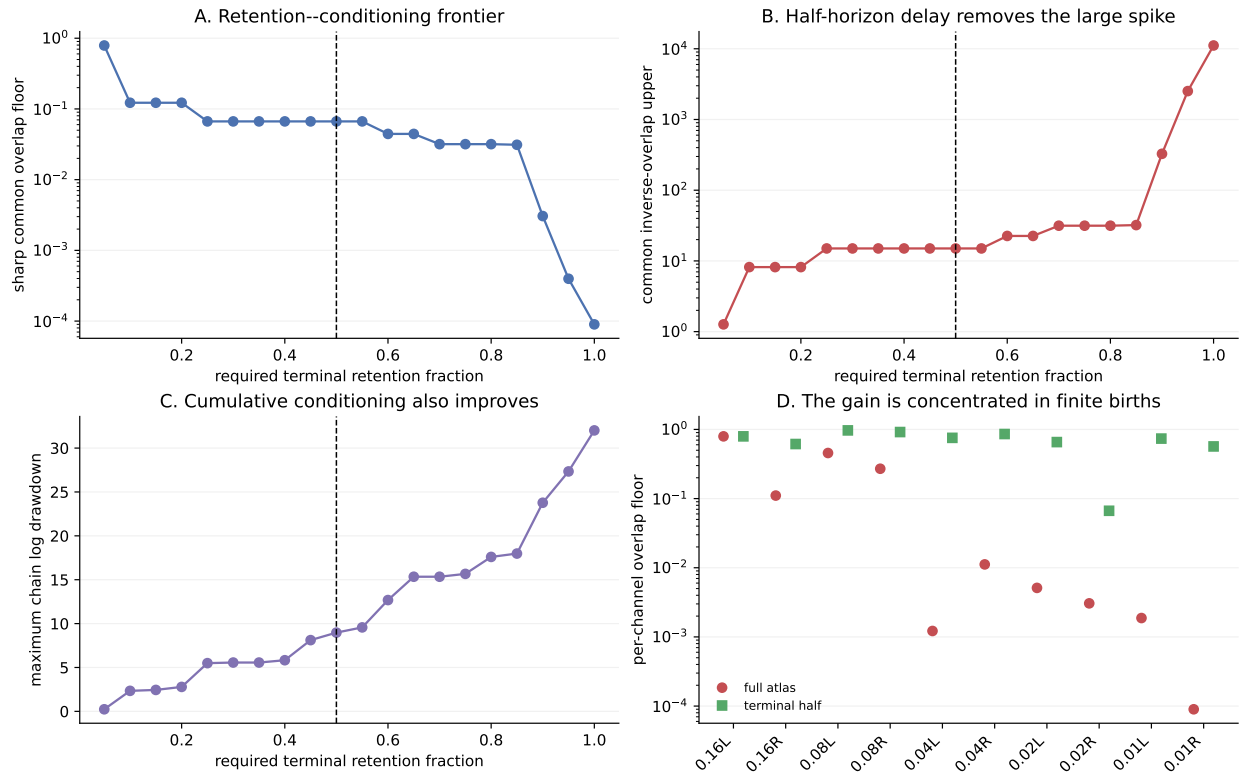


Figure 1: The exact retention frontier, inverse bound, cumulative drawdown, and per-channel improvement from the terminal-half reset.